

Bottle Detection Evaluation Report

Name: Ayesha Imran

Project: Bottle Detection using YOLOv8

Framework: Ultralytics YOLOv8

Language: Python

1. Introduction

This project aims to detect bottles using the YOLOv8 object detection algorithm. The model was trained on a custom dataset collected and annotated using Roboflow. The final model can detect bottles from images, videos, and live webcam input.

2. Dataset

Dataset Source:

Roboflow

Classes:

Bottle

Number of Images:

837

Train / Validation / Test Split:

70/10/2

3. Training Configuration

Model:

YOLOv8n

Epochs:

50

Image Size:

640 × 640

Batch Size:

8

Optimizer:

Auto

Device:

CPU

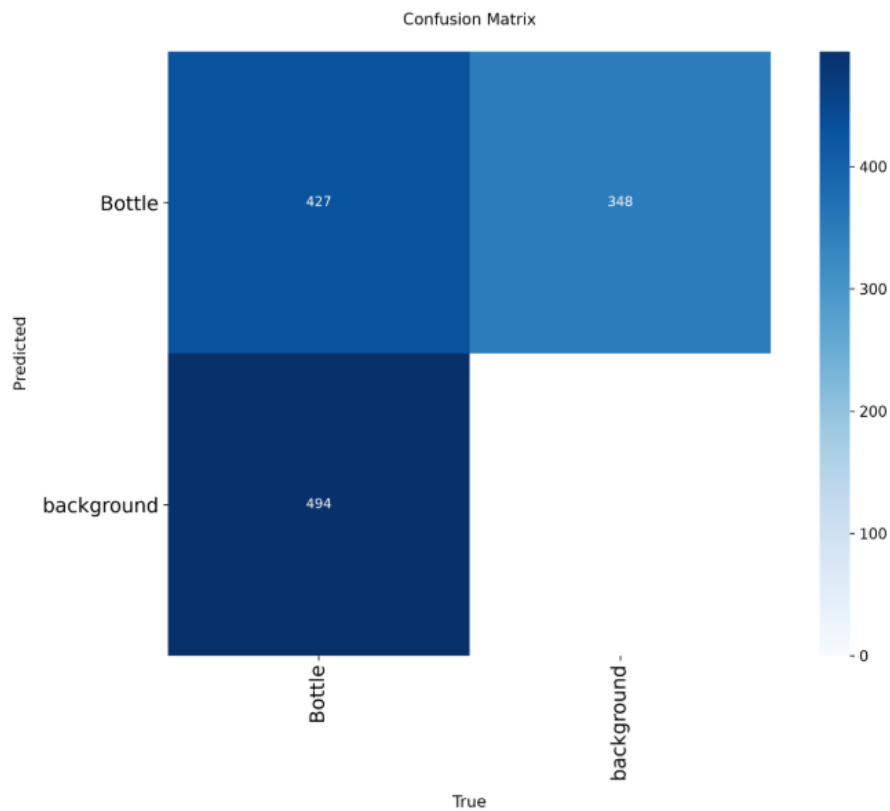
Framework:

PyTorch

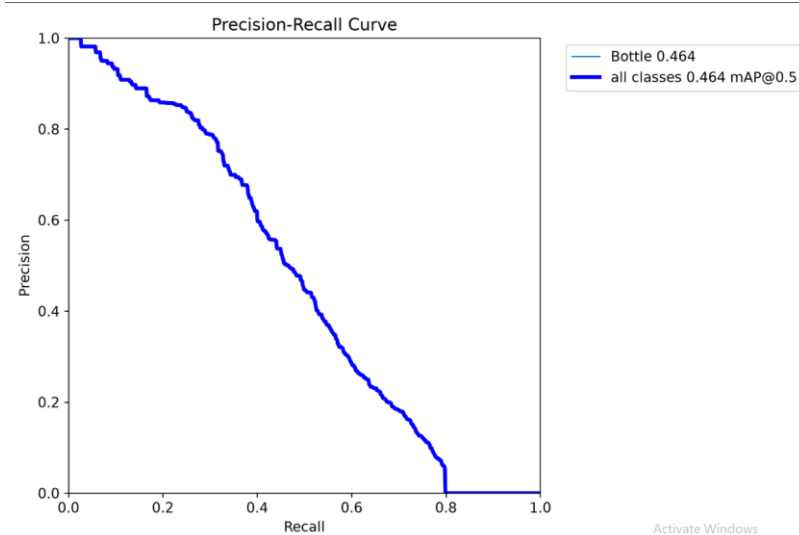
4. Evaluation Metrics

Metric	Result
Precision	0.575
Recall	0.414
mAP50	0.464
mAP50-95	0.271

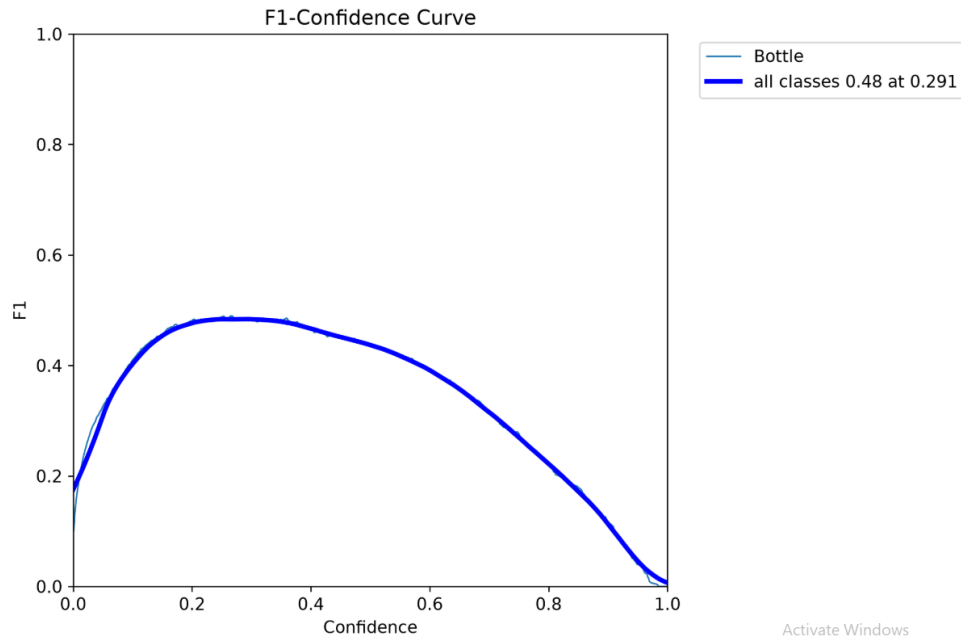
5. Confusion Matrix



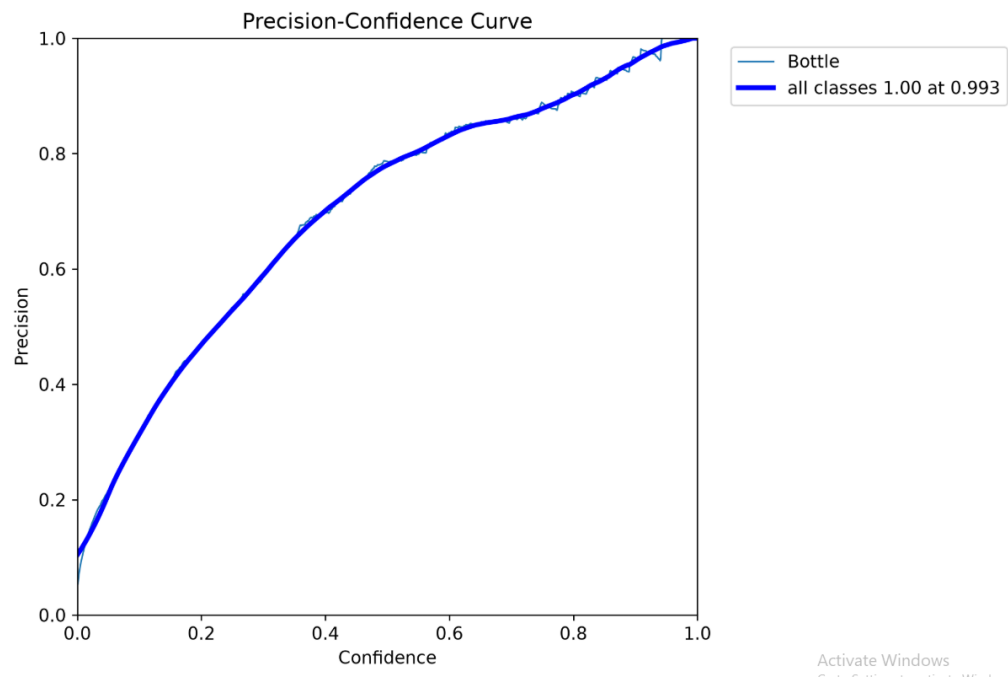
6. Precision-Recall Curve



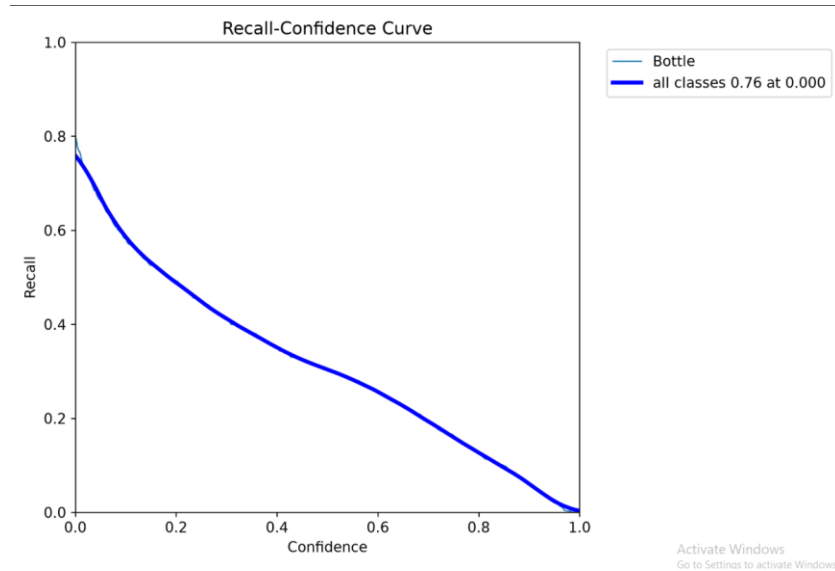
7. F1 Curve



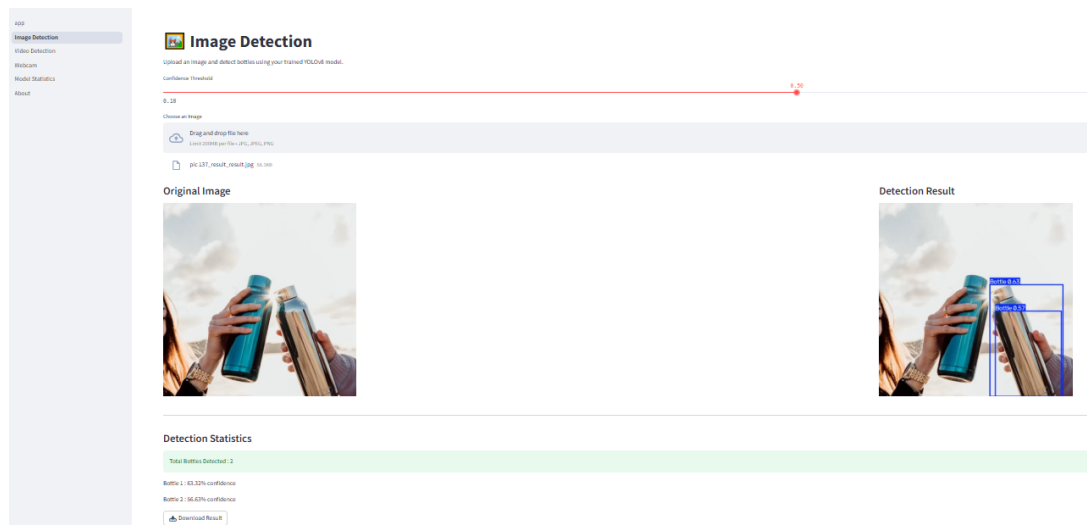
8. Precision Curve



9. Recall Curve



10. Sample Predictions



11. Strengths

- Detects bottles accurately in good lighting.
- Works on images, videos, and webcam.
- Fast inference using YOLOv8.
- Professional Streamlit dashboard.
- Easy to use interface.

- Supports confidence-based detection.

12. Limitations

- Accuracy decreases under poor lighting.
- Small or heavily occluded bottles may not be detected.
- CPU inference is slower than GPU inference.
- Performance depends on dataset quality and diversity.

13. Future Improvements

- Train using YOLOv8s or YOLOv8m.
- Increase dataset size.
- Add multiple object classes.
- Deploy as a web application.
- Optimize for real-time GPU inference.
- Add object tracking for video streams.

14. Conclusion

The Bottle Detection System successfully demonstrates the use of deep learning for real-time object detection. The trained YOLOv8 model can detect bottles from multiple input sources and provides satisfactory performance for educational and portfolio purposes. Future improvements can further increase accuracy and deployment readiness.